- The negative of a matrix $A_{m\times n}$ denoted by $-A_{m\times n}$ is the matrix formed by replacing each element in the matrix $A_{m\times n}$ with its additive inverse.

- **Addition and subtraction of matrices**

  Two matrices can be added or subtracted if they have the same order. To add or subtract two matrices, simply add or subtract the corresponding elements.

- **Multiplication of matrix by a scalar**

  We can multiply the elements of the given matrix A by a non-zero number k to obtain a new matrix kA whose elements are multiplied by k. The matrix kA is called scalar multiplication of A.

  Thus if $A=(a_{ij})_{m\times n}$ then , $kA=(ka_{ij})_{m\times n}$ for all $i = 1,2,\ldots,m$ and for all $j = 1,2,\ldots,n$.

ICT CORNER

ICT 3.1

Step 1: Open the Browser type the URL Link given below (or) Scan the QR Code. Chapter named **"Algebra"** will open. Select the work sheet **"Simultaneous equations"**

Step 2: In the given worksheet you can see three linear equations and you can change the equations by typing new values for a, b and c for each equation. You can move the 3-D graph to observe. Observe the nature of solutions by changing the equations.

Step 1



Step 2



Expected results

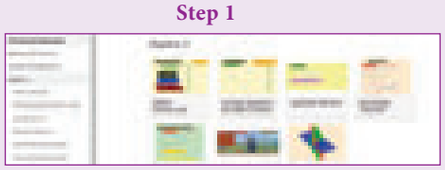


ICT 3.2

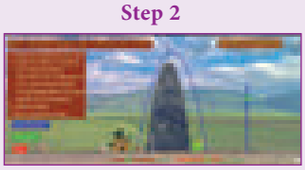
Step – 1: Open the Browser type the URL Link given below (or) Scan the QR Code. GeoGebra work book named "ALGEBRA" will open. Click on the worksheet named "Nature of Quadratic Equation".

Step - 2: In the given worksheet you can change the co-efficient by moving the sliders given. Click on "New position" and move the sliders to fix the boundary for throwing the shell. Then click on "Get Ball" and click "fire" to hit the target. Here you can learn what happen to the curve when each co-efficient is changed.

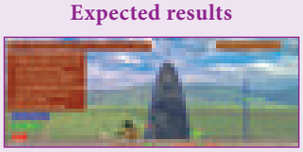
Step 1



Step 2



Expected results



You can repeat the same steps for other activities

https://www.geogebra.org/m/jfr2zzgy#chapter/356193
or Scan the QR Code.

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